UVA DS2003 Communicating with Data
Fall 2026

Storytelling with Data
Narrative Visualization

Sarah Groves
based on materials by Hannah Bako and a lot of other people

Project 2: Interactive Notebook & Infographic/Science Visualization

- *Explanatory analysis:* together, your notebook and final deliverable will tell a story about a dataset
- This will require you to:
  - First *explore* the data (a short EDA)
  - Determine what story you are interested in telling
  - Put together *interactive* visualizations into a notebook *report*
  - Pick a single message you will emphasize in a single graphic (infographic or scivis)
  - Present the notebook and graphic to the class
- Unlike the first project, where you asked questions of the dataset and reported what you found, you will need to identify a key message to persuade the audience based on your analysis of the data
  - This is not a modeling class, so no sophisticated modeling is necessary. However, your analysis should lead to a primary argument you want to make that is sufficiently supported by the data

Project 2: deadlines

- This **Wednesday**: Pick your groups (3 people per group) by 10am on Wednesday; at 10, I will finish sorting the groups if you don't have a preference for who you work with
- Oct 25: Project Proposal
- Nov 15: Narrative Structure Outline
- Nov 22: Design critique reflection
- Dec 1: Final presentation: interactive walkthrough and graphic presentation
- Dec 7: Final interactive notebook
- Dec 7: Final infographic/science visualization and design rationale

Explanatory visualizations: outline of the second half of the semester

- Storytelling
- Interactive design: plotly and ipywidgets
- Non-tabular data: networks and trees
- Infographics
- Science visualizations
- Data journalism
- Data quality, ethics and rhetoric

What you will learn today

The role of storytelling in data communication

How to identify the components of a data story

How to make the same mistakes others have made in telling story with data

Exploratory vs. explanatory visualization

- Exploratory visualization is for *you*: many quick, rough charts to understand the data and find what is interesting
- Explanatory visualization is for *your audience*: a designed chart that communicates one specific message to specific people
- Exploratory is finding the story; explanatory is telling it
- Project 1 was exploratory. Project 2 is explanatory.

Narrative Story-Telling

narrative (n): An account of a series of events, facts, etc., given in order and with the establishing of connections between them.

Effective story-telling “require[s] skills like those familiar to movie directors, beyond a technical expert’s knowledge of computer engineering and science.” - Gershon & Page ‘01

Think | Pair | Share

- What’s your favorite movie or book?
- Why is it your favorite?
- How does it go beyond a simple chronological “what happened” to tell a story that moved you?

Movies and books are “author-driven”… there is a strong narrative with little choice for the reader

Interacting with raw information is “reader-driven”… open interaction with no predefined story

The sweet spot: narrative visualization balances your intended narrative path (as the designer) with interactive reader exploration.

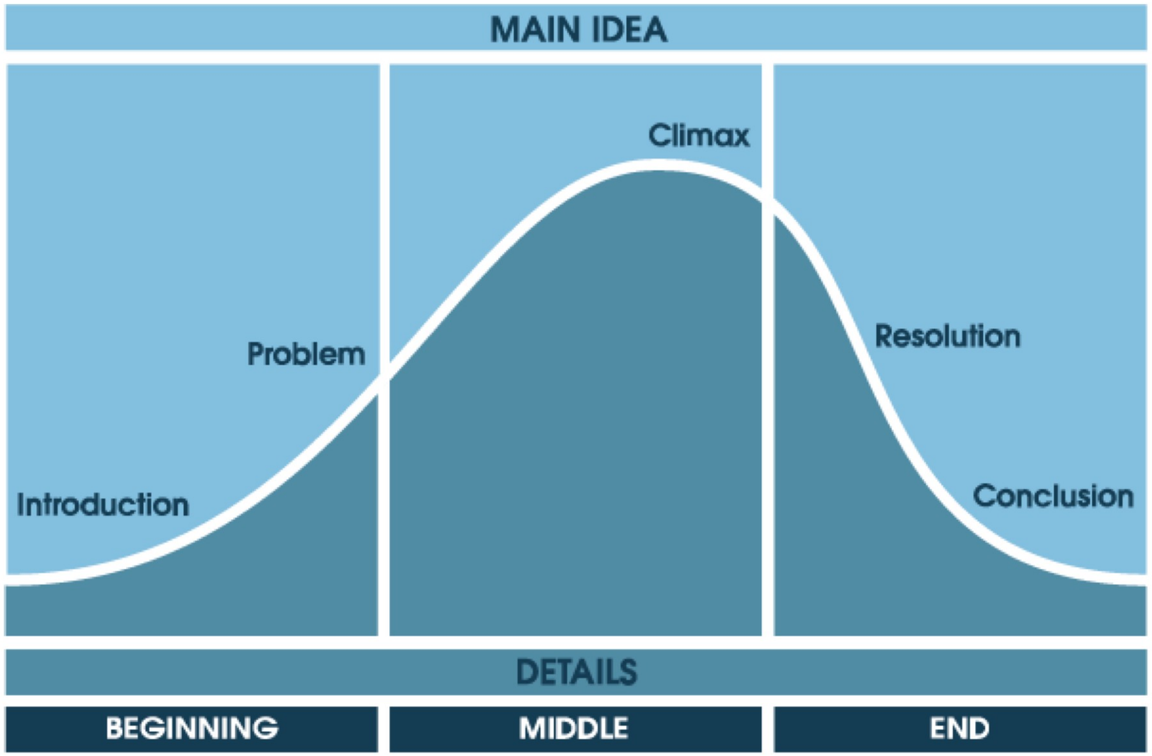
But *how* do we tell a good story with data?

Storytelling

Good stories do more than provide facts and data

- they situate and give context
- they engage
- they educate

Who / What / Where / When
Why / How



© 2016 Clever Prototypes, LLC

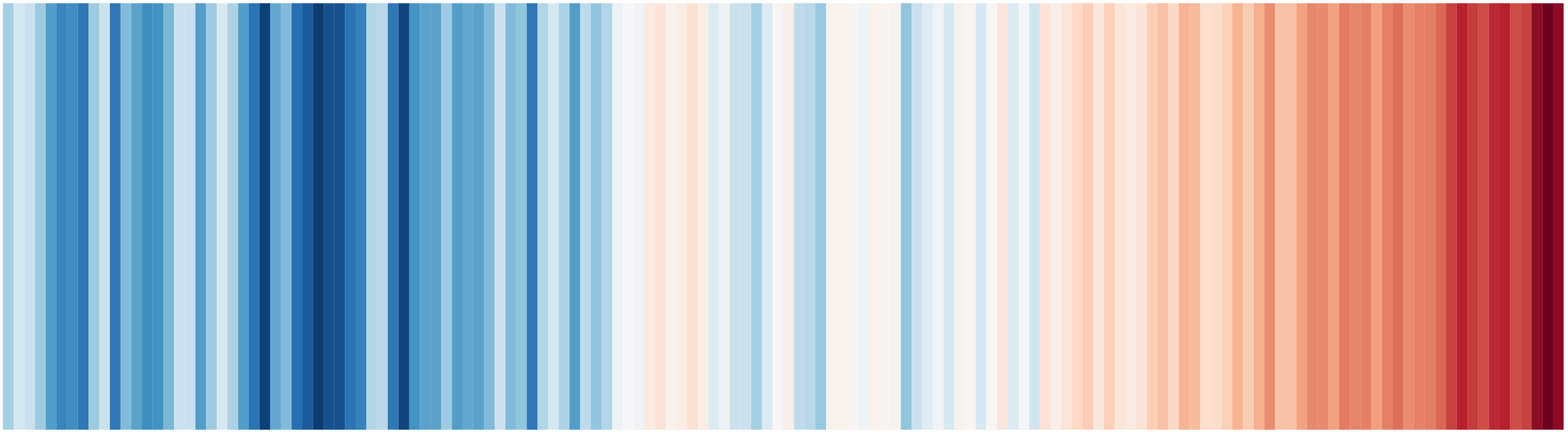
Storytelling with data: simple, powerful messaging

Underscore your arguments with Data/Facts

Leverage the power of Visualization

- Show Trends
- Show Correlations
- Show Outliers
- Convey Magnitudes

Storytelling with data: simple, powerful messaging



1880 2025

Trend

Blue gives way to red: no axis needed to see the direction

Outliers

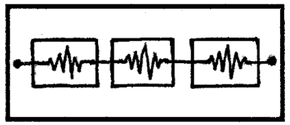
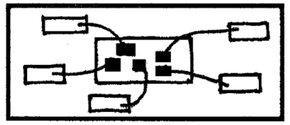
The 2023–24 stripes are darker than anything before them

Magnitude

Darker means farther from normal: 2024 ran 1.3 °C above the 1951–80 average

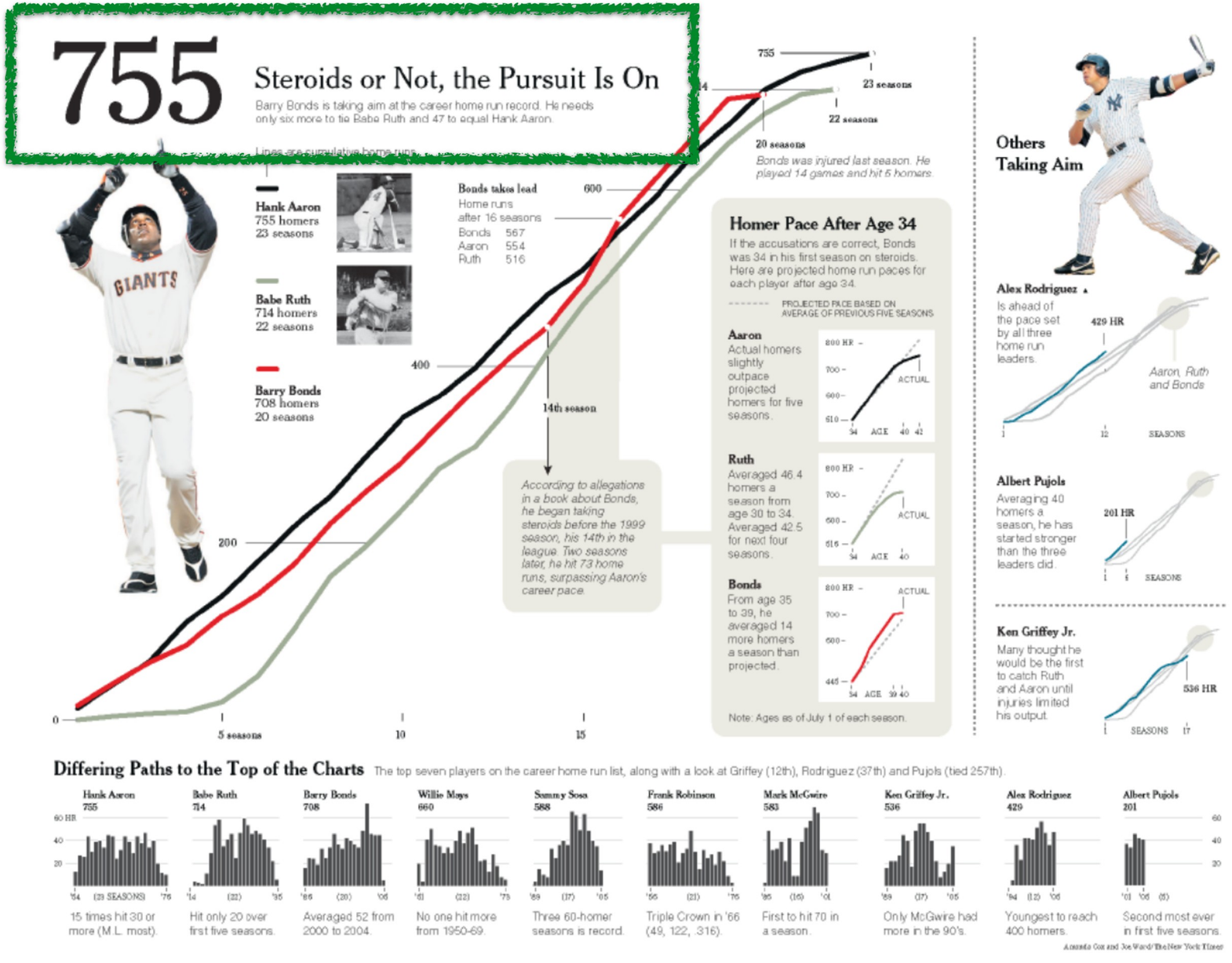
Data: NASA GISTEMP v4, global surface temperature anomaly; white = 1951–1980 average. Design after Ed Hawkins' "warming stripes" (#ShowYourStripes, 2018).

Like a film director, you must direct the viewer's attention to your story… while making sure you tell the truth about the data.

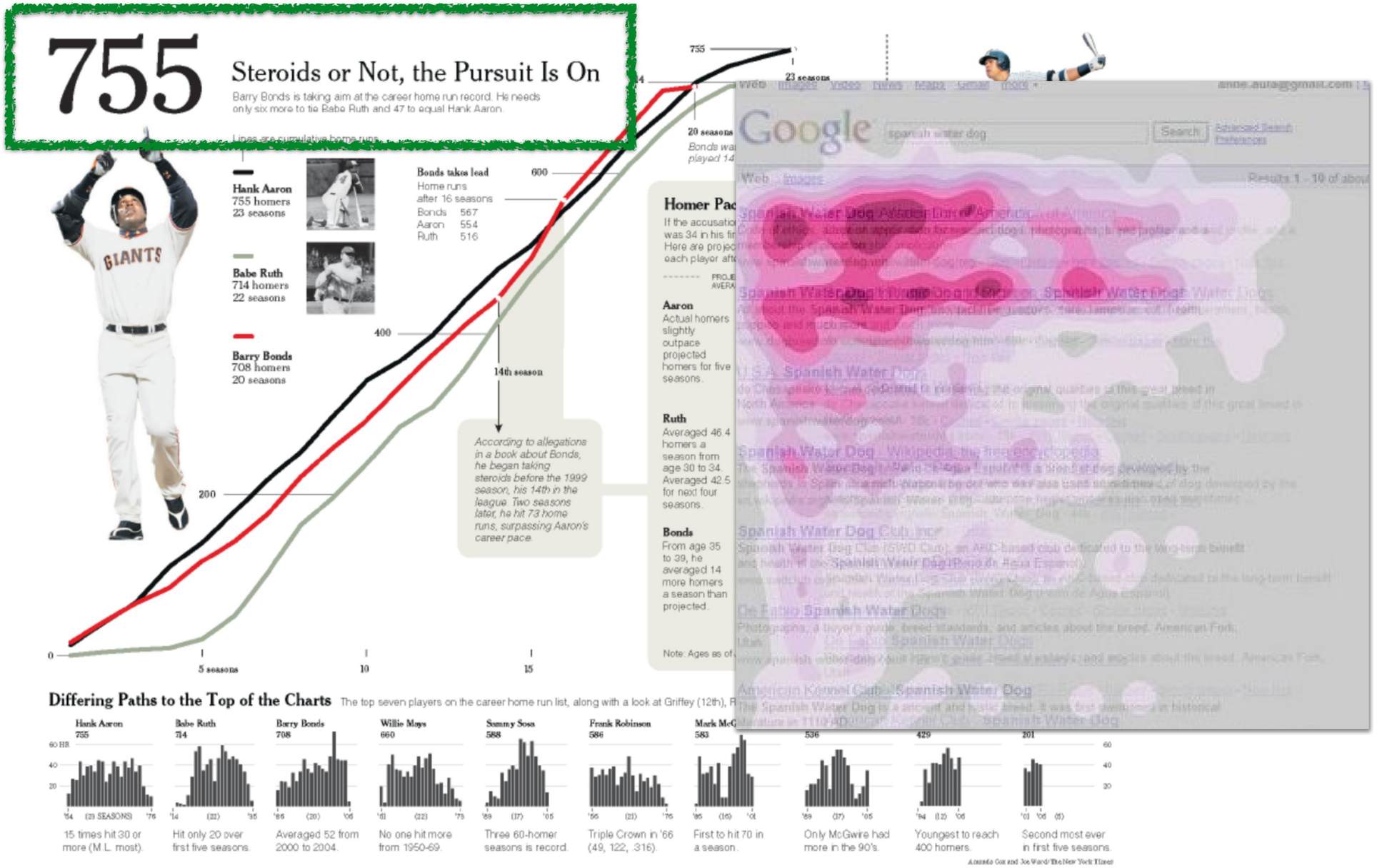
- You must first do EDA to see what the data supports
- Just like your first project, you will find a few interesting conclusions from the data; things that would might want to communicate to others!
- These ideas become the core of the story you want to tell
- Lead the viewer where you want them to go, giving a little freedom to explore
  - Martini glass model for narrative structure 
  - Interactive progression through charts 
  - Drill-down story 

Components of a good data story

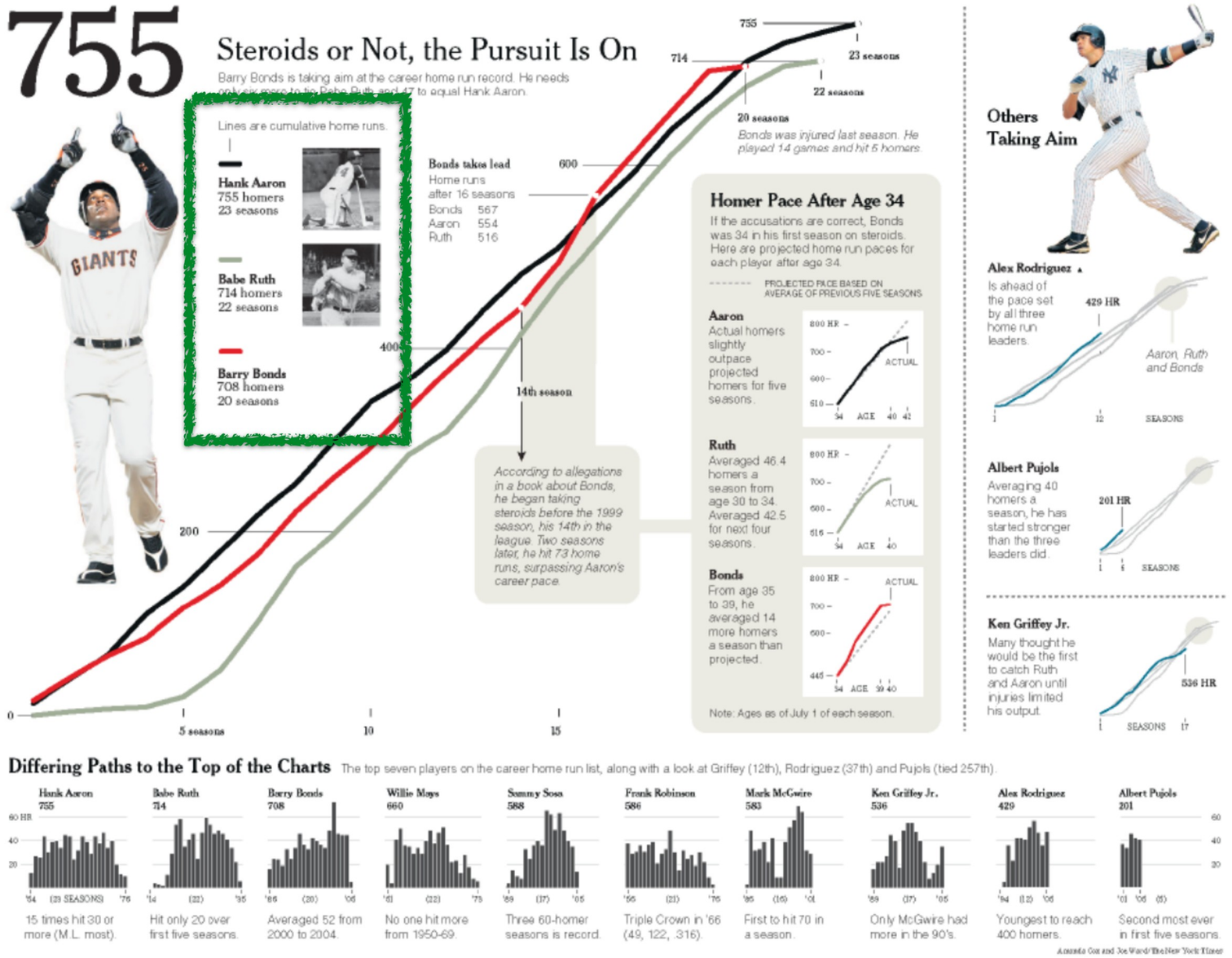
Introduction



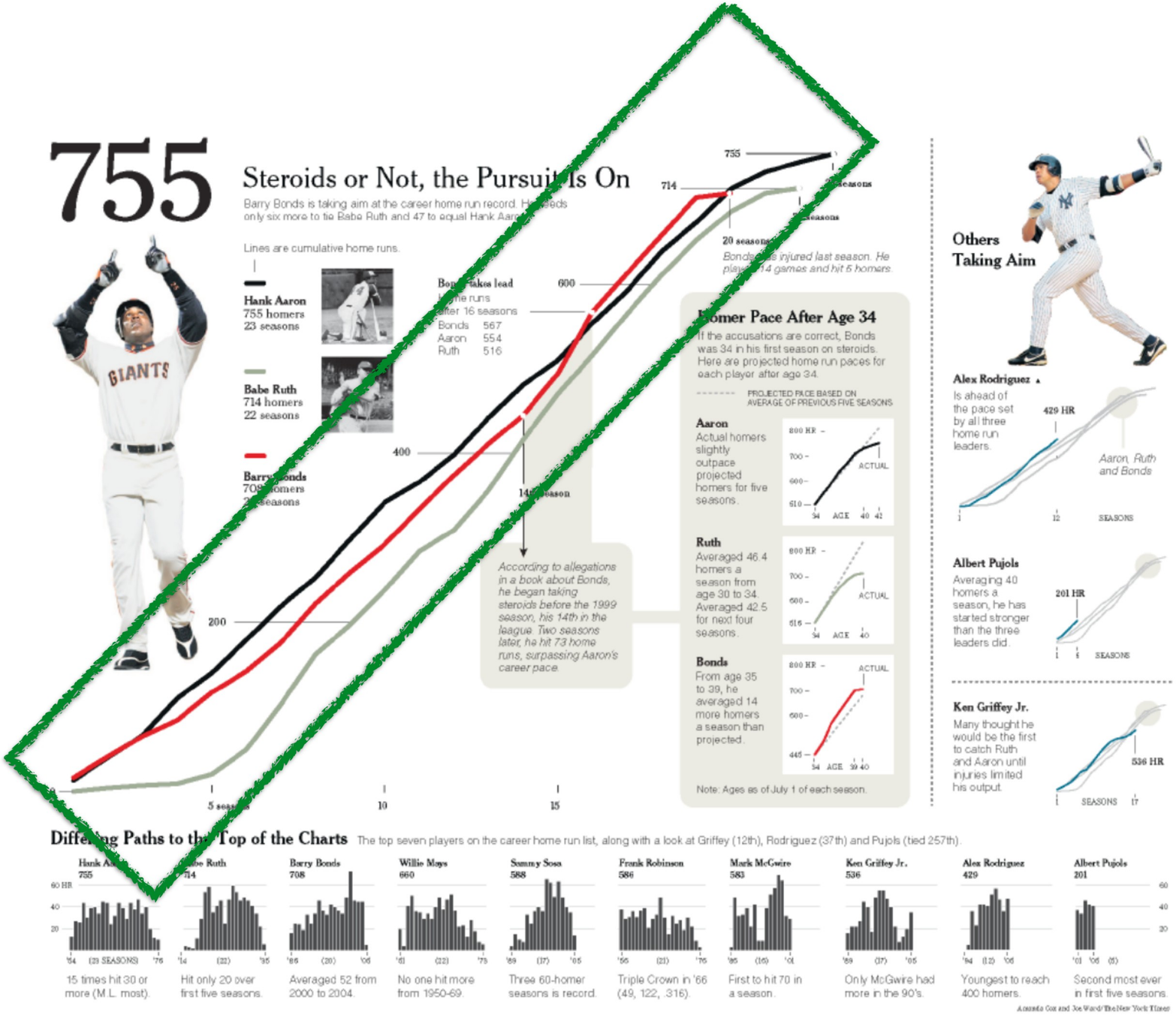
Introduction



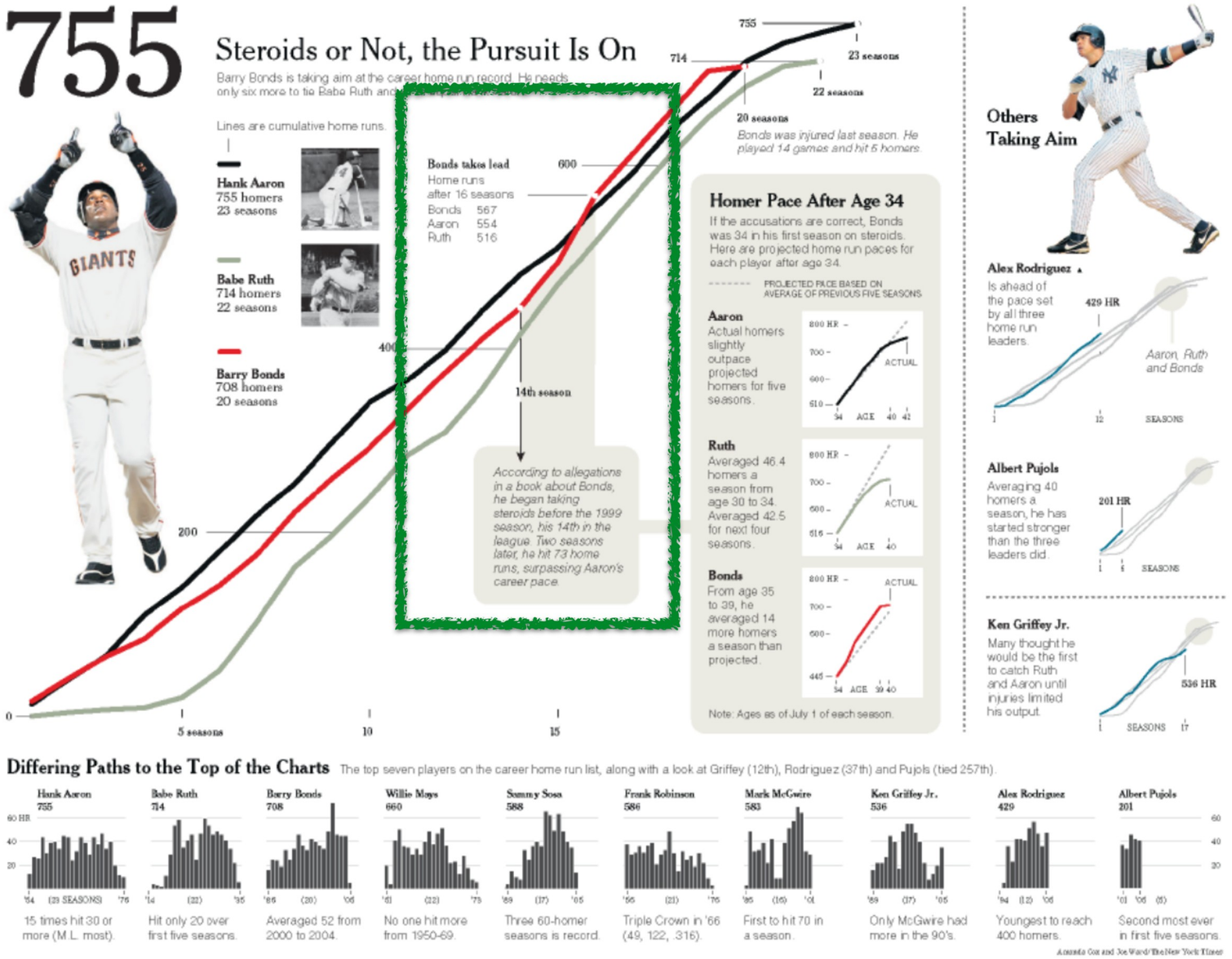
Context / Legend



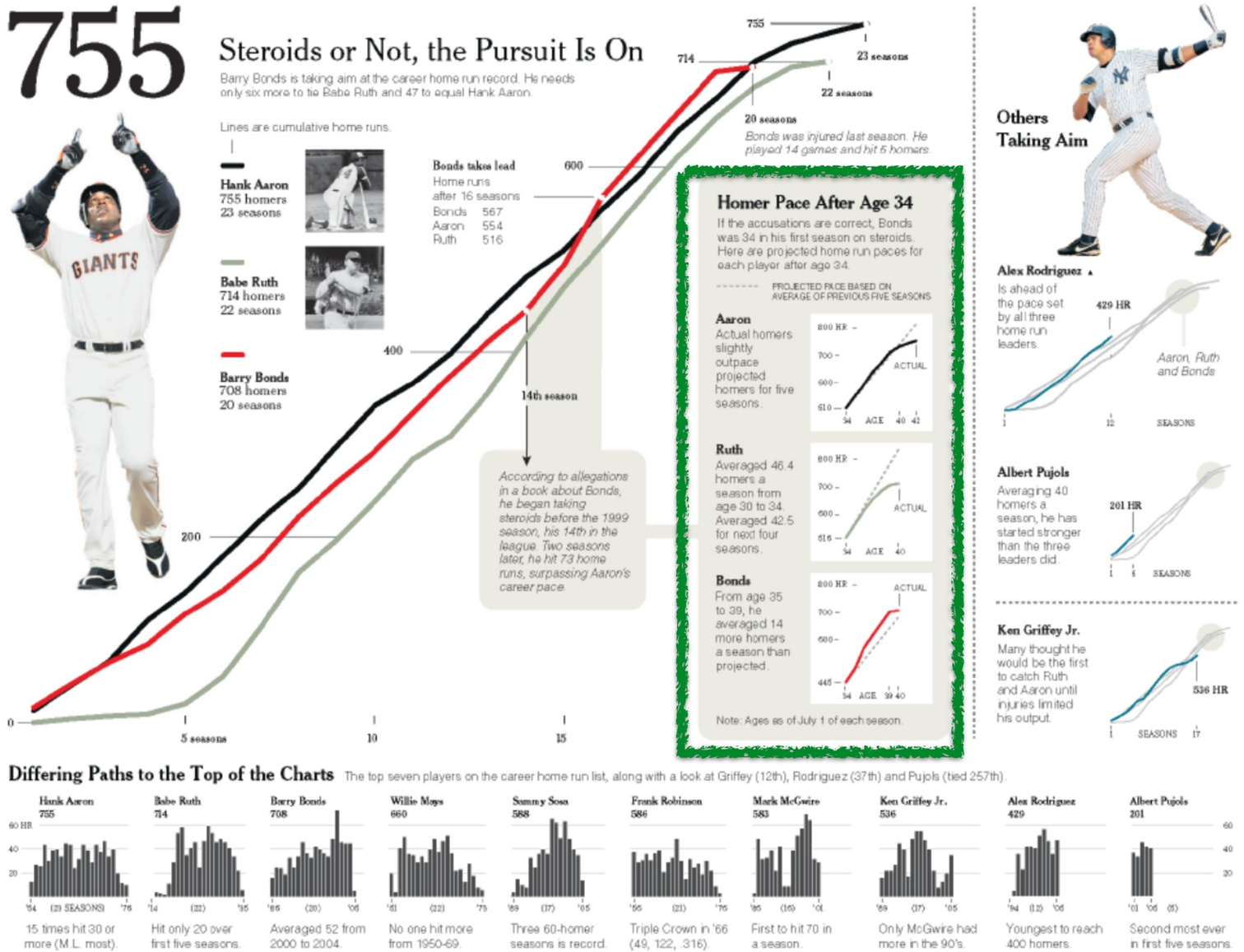
Main Story


755
Steroids or Not, the Pursuit Is On
Barry Bonds is taking aim at the career home run record. He needs only six more to tie Babe Ruth and 47 to equal Hank Aaron.
Lines are cumulative home runs.
Hank Aaron
755 homers
23 seasons
Babe Ruth
714 homers
22 seasons
Barry Bonds
708 homers
22 seasons
Bonds takes lead
home runs
after 16 seasons
Bonds 567
Aaron 554
Ruth 516
755
714
600
400
200
20 seasons
Bonds was injured last season. He played 14 games and hit 5 homers.
According to allegations in a book about Bonds, he began taking steroids before the 1999 season, his 14th in the league. Two seasons later, he hit 73 home runs, surpassing Aaron's career pace.
5 seasons
10
15
Homer Pace After Age 34
If the accusations are correct, Bonds was 34 in his first season on steroids. Here are projected home run paces for each player after age 34.
PROJECTED PACE BASED ON AVERAGE OF PREVIOUS FIVE SEASONS
Aaron
Actual homers slightly outpace projected homers for five seasons.
Ruth
Averaged 46.4 homers a season from age 30 to 34. Averaged 42.5 for next four seasons.
Bonds
From age 35 to 39, he averaged 14 more homers a season than projected.
800 HR
700
600
510
445
ACTUAL
AGE
Note: Ages as of July 1 of each season.
Others Taking Aim
Alex Rodriguez
Is ahead of the pace set by all three home run leaders.
429 HR
Aaron, Ruth and Bonds
SEASONS
Albert Pujols
Averaging 40 homers a season, he has started stronger than the three leaders did.
201 HR
Ken Griffey Jr.
Many thought he would be the first to catch Ruth and Aaron until injuries limited his output.
536 HR
Differing Paths to the Top of the Charts
The top seven players on the career home run list, along with a look at Griffey (12th), Rodriguez (37th) and Pujols (tied 257th).
Hank Aaron 755
Babe Ruth 714
Barry Bonds 708
Willie Mays 660
Sammy Sosa 588
Frank Robinson 586
Mark McGwire 583
Ken Griffey Jr. 536
Alex Rodriguez 429
Albert Pujols 201
60 HR
40
20
(23 SEASONS)
15 times hit 30 or more (M.L. most).
Hit only 20 over first five seasons.
Averaged 52 from 2000 to 2004.
No one hit more from 1950-69.
Three 60-homer seasons is record.
Triple Crown in '66 (49, 122, .316).
First to hit 70 in a season.
Only McGwire had more in the 90's.
Youngest to reach 400 homers.
Second most ever in first five seasons.
Amanda Cox and Joe Ward/The New York Times

Annotation of key point: a change in trends



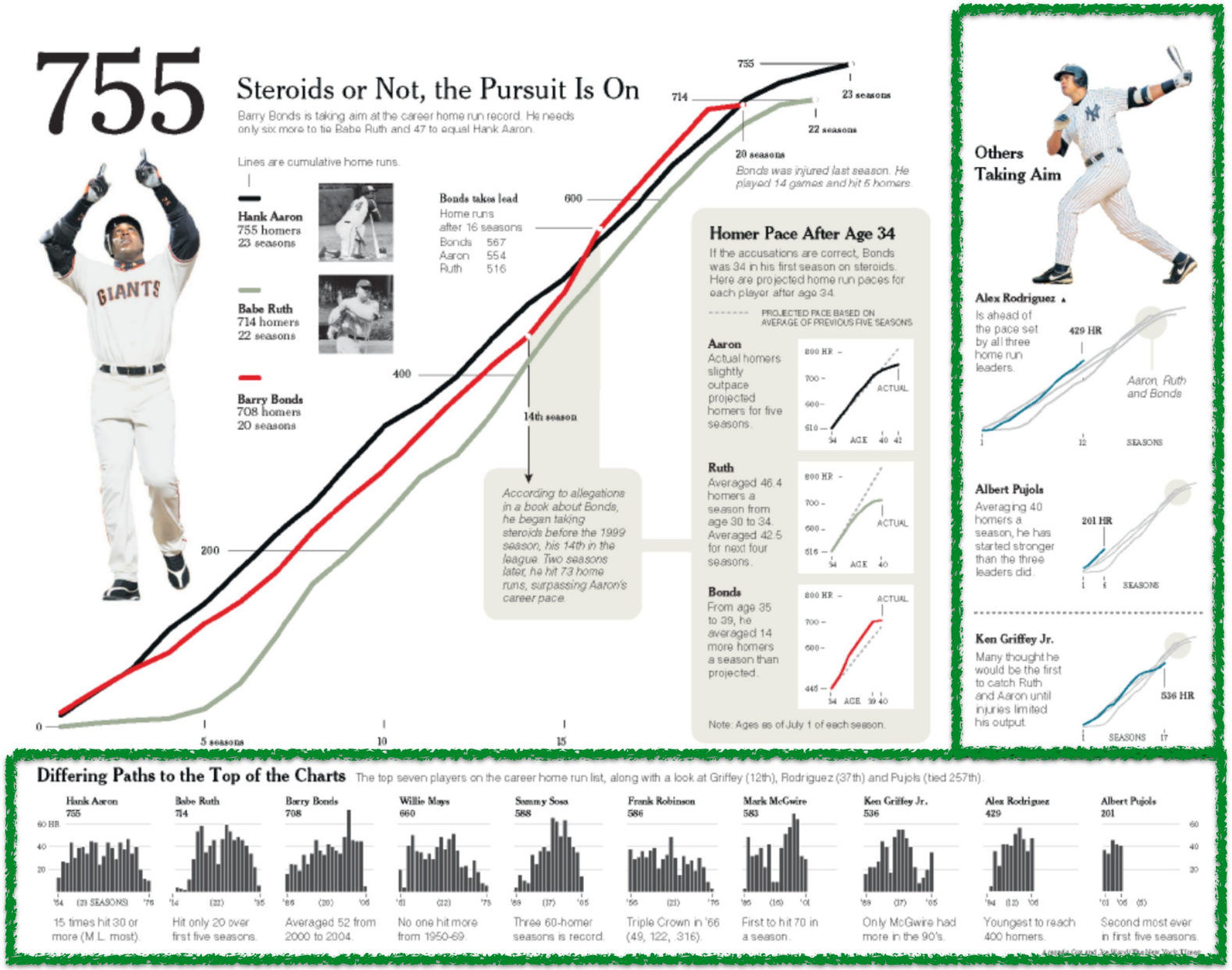
Context, using matching on content (color, trend)



Multi-Messaging - in-depth information

Other Players

Different Paths



Where failed data stories go wrong

Think | Pair | Share

- Think of a time someone tried to tell your or others a story with data, and it failed.
- What do you think went wrong?
- Why was the intended message not communicated effectively?

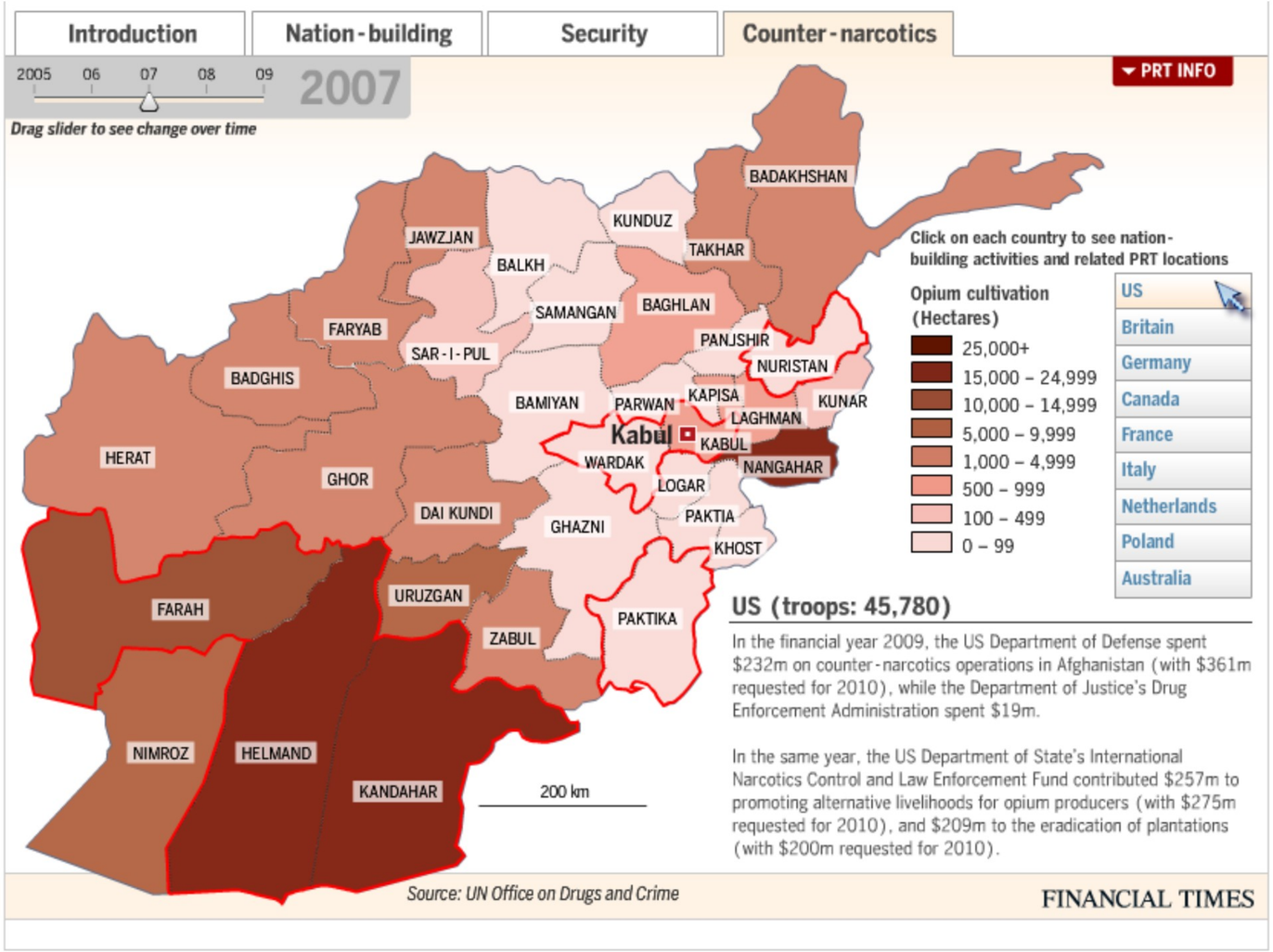
Afghanistan: behind the front line

Intended purpose of the graphic: to establish indicators of success by which to evaluate the development work being done in Afghanistan

- Semantically consistent color

- Details on demand

- However, it does not guide the viewer through the data at all → too much flexibility → viewer can't reach meaningful conclusions intended by the designers



The graphic "suffers by putting exploratory power into the hands of the viewer without sufficient guidance"

Minnesotan Employment Explorer

Context: in July 2008, Minnesota was in recession, so Minnesota Public Radio (MPR) released “Minnesota Slowdown:” stories detailing the state’s economic malaise and shares interviews with residents hit hard by the downturn.

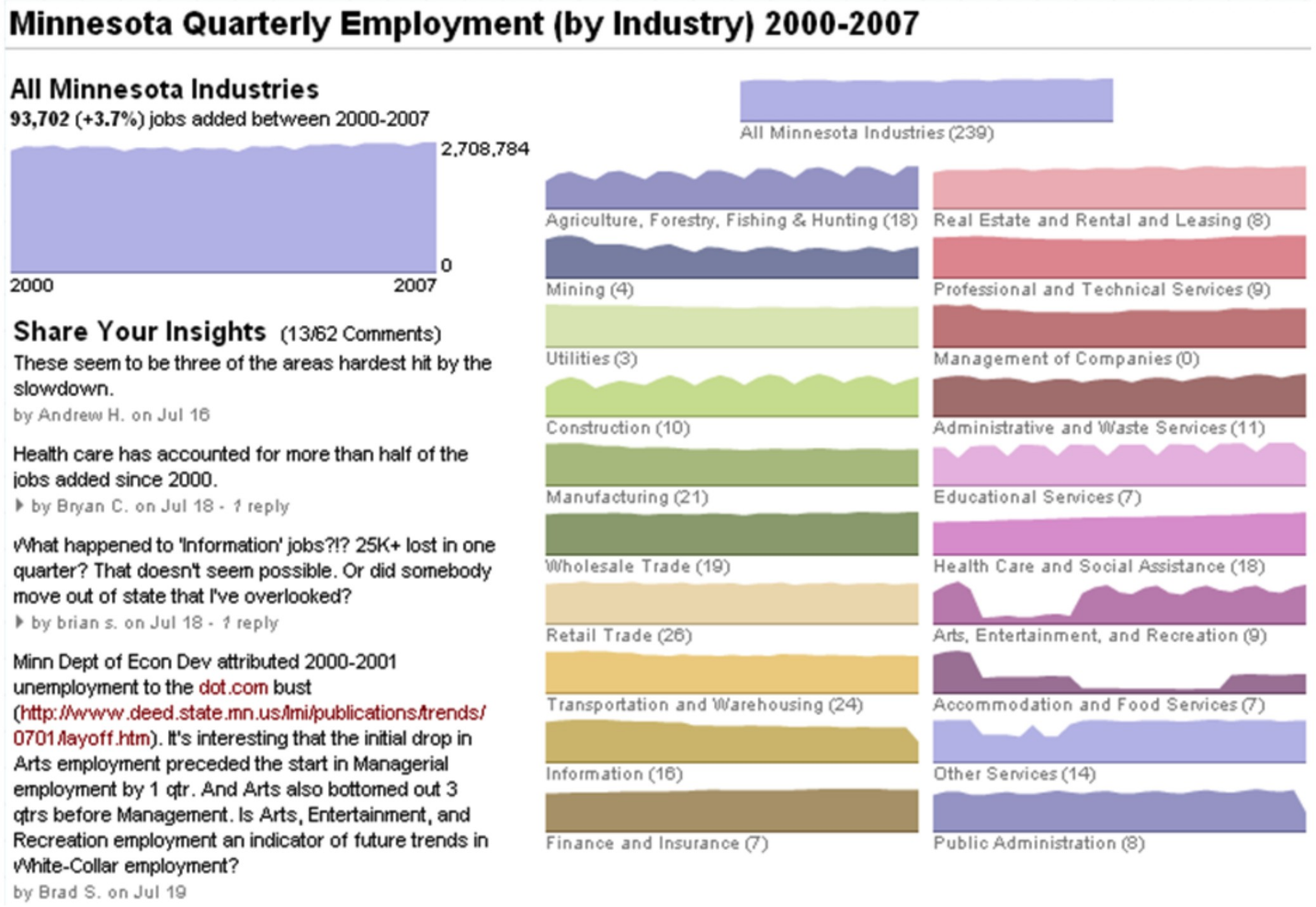
Intended purpose: engage readers in finding and telling their own stories in the data.

Only 23 people engaged, and not Minnesotans

- No tutorial and too much data → confused users

- No rich, emergent story

- Disconnected from the narrative (graphic showed 2000-2007)



Summary

- Explanatory visualization is about communicating a story that you found in the data
  - It is not about manipulating the data to tell a story that you already think is true!
  - Effective storytelling should guide the viewer to the core message and allow for a restrained amount of exploration
  - In Western cultures, we read left to right and top to bottom, so most important information should be top left
- Use clear annotations and legends to highlight the main story
- Too much data/chronological “as you made them” charts will not be effective